Measuring and Modeling Perceived Humanness in AI-Generated Political Discourse

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Perceived Humanness: The extent to which a post exhibits natural, contextually appropriate, and stylistically nuanced characteristics associated with human writing.

Research Questions

1. Can people distinguish between human- and AI-generated social media posts?
2. What stylistic features drive perceptions of humanness?
3. Can we fine-tune chatbots to sound more human-like?

Motivation

Political social media content plays a major role in shaping opinion. AI language models are now part of this online discourse. But how much of the conversation is perceived to be human?

Example Labeling Task:

Which of the following two X Posts is more likely to be generated by an AI Chatbot?

1. "why do politicians always avoid direct answers?? 😒 #Debatenight" - X Post
2. Tonight's #debate will air live on ABC and can be live-streamed on ABC News Live, #Disney+ and #Hulu at 9 p.m. EDT. #DebateNight https://t.co/xzfcgwsr - ChatGPT 4o

Acknowledgements

Thank you to Lisa, Leticia, Tiago, Sejin, Becky, Autumn and Kenny for an amazing opportunity and year of collaboration, support, and growth. Thank you to MDI, the McCourt School and the Tech and Public Policy Grants Program for supporting this work.

Study 1: Assessing Perceived Humanness

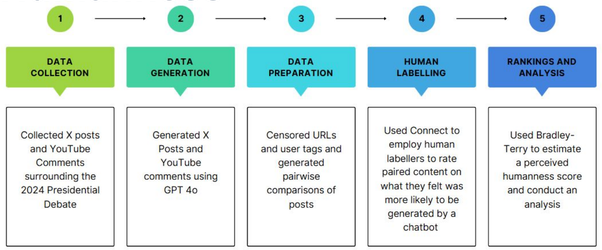


Figure 1: Data Collection Methodology

Ranking Perceived Humanness

- Labellers could generally distinguish between AI-generated and human-authored content
- Separation on X is more than on YouTube

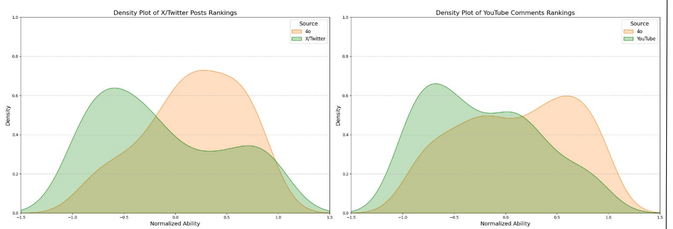


Figure 2: Density plot of PHI scores by model type, showing human posts often outscored fine-tuned models.

Affectability Results

- Positive sentiment often perceived as AI-authored.
- Negative sentiment and offensive language linked to human authorship.

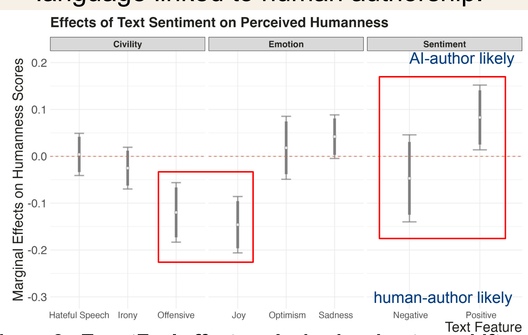


Figure 3: TweetEval affect analysis showing tone shifts across perceived humanness ranks

Study 2: Fine-Tuning with Perceived Humanness

Using data from Study 1 allowed us to model perceived humanness in 3 ways:

Method	Training Data			Used For
1	Prompt	Chosen	-	Supervised Fine Tuning
2	Prompt	Chosen	Rejected	Direct Preference Optimization Fine Tuning
3	Post	PHI-Score	-	PHI-Predict Regression Tool

Key Findings

➔ **More tuning ≠ more human:** Adding preference data often reduced humanness; DPO altered style but didn't improve perception.

➔ **Perception is subjective:** Annotator judgments varied, and text features alone weakly predicted humanness.

➔ **Smaller, cleaner data helps:** A DPO subset trained only on human-preferred pairs produced more natural outputs.



Figure 4: Human Evaluation of Fine-Tuned Models

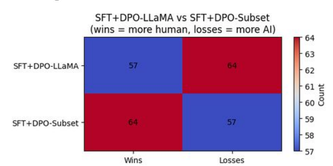


Figure 5: DPO-Subset Outperforms DPO in Human Judgments
The DPO-Subset model was trained only on comparisons where **human-authored posts were preferred over AI**, providing a clearer training signal.

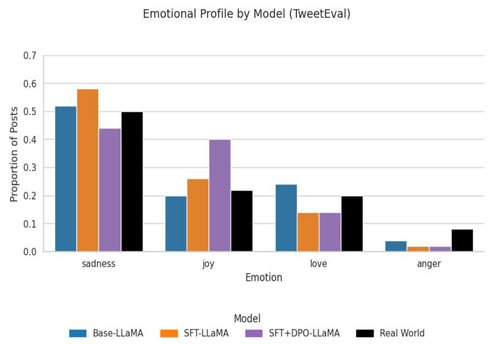


Figure 6: Emotional profile of model outputs based on TweetEval.
All models skewed toward sadness, with SFT+DPO generating the most joy. Anger was rarely expressed across all systems, suggesting limited emotional range in that category.

Limitations

- Subjective judgments: Annotators often disagreed on what felt human, and platform context (X vs. YouTube) influenced perception.
- Fine-tuning tradeoffs: More data didn't always improve humanness.
- Weak predictions: Text-based models couldn't reliably estimate perceived humanness.

Conclusion

Humanness is complex: tone, emotion, and style all matter.

More tuning ≠ more human: DPO sometimes reduced authenticity.

DPO-Subset helped: Smaller, cleaner data improved outputs.

Next steps:

1. Add more platforms (TikTok, TruthSocial)
2. Improve annotation quality (doublel-labeling)
3. Explore RLHF and adaptive sampling to build models

Ethical Considerations

Improving the humanness of AI-generated text may make it harder to distinguish from real content, increasing the risk of misinformation. Our work aims to better understand and control how AI outputs are perceived to support transparency and responsible use.